

**PREDICTIVE MODELLING OF AIR QUALITY:
PERFORMANCE COMPARISON OF SUPERVISED
MACHINE LEARNING, DEEP LEARNING, AND
EXTREME LEARNING MACHINES**

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PREDICTIVE MODELLING OF AIR QUALITY: PERFORMANCE COMPARISON OF SUPERVISED MACHINE LEARNING, DEEP LEARNING, AND EXTREME LEARNING MACHINES

*A Project Report Submitted in partial fulfillment of the
requirements for the award of the degree of*

**Bachelor of Technology
in
Computer Science & Engineering**

by

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ABSTRACT

KEYWORDS: Air Quality Index (AQI), Air Pollution Forecasting, Supervised Machine Learning, Deep Learning, Extreme Learning Machines (ELM), Genetic Algorithms (GA), Decision Trees, Support Vector Machines, Logistic Regression, Random Forest, Naïve Bayes, K-Nearest Neighbors, SO₂, NO₂, RSPM, SPM, PM_{2.5}.

A comprehensive analysis comparing the performance of supervised machine learning, deep learning, Extreme Learning Machines (ELM), and Genetic Algorithm (GA) models in predicting the Air Quality Index (AQI). With growing concerns about air pollution and its severe impact on both public health and the environment, developing accurate and timely forecasting models has become critical for effective intervention strategies. This study leverages real-world datasets to assess the predictive accuracy, computational efficiency, and applicability of these different models in various forecasting scenarios. We focus on key attributes such as sulfur dioxide (SO₂), nitrogen dioxide (NO₂), respirable suspended particulate matter (RSPM), suspended particulate matter (SPM), and PM_{2.5} to predict AQI. The results indicate that deep learning models excel in predictive accuracy, while ELM models offer an optimal balance between accuracy and computational speed, making them suitable for real-time applications. Genetic Algorithms, known for their optimization capabilities, also demonstrate promise in refining model parameters, further improving the forecasting performance. Additionally, supervised machine learning techniques, including decision trees, support vector machines, logistic regression, random forest, naïve Bayes, and k-nearest neighbors, continue to show reliable results, underscoring their importance in environmental forecasting tasks. These findings provide a comparative perspective, aiding in the selection of appropriate models for air quality prediction depending on specific requirements for accuracy, efficiency, and scalability.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
CPCB	Central Pollution Control Board
SPCB	State Pollution Control Board
AQI	Air Quality Index
PM	Particulate Matter
RSPM	Respirable Suspended Particulate Matter
SPM	Suspended Particulate Matter
SO ₂	Sulfur Dioxide
NO ₂	Nitrogen Dioxide
DL	Deep Learning
ML	Machine Learning
GA	Genetic Algorithm
DFD	Data Flow Diagram
ELM	Extreme Learning Machine
SVM	Support Vector Machine
KNN	K-nearest Neighbor

CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

Air pollution represents a significant environmental issue of the 21st century, severely affecting human health, ecosystems, and global climate systems. The rapid industrial growth and urban expansion worldwide have exacerbated the release of harmful pollutants, including particulate matter (PM_{2.5} and PM₁₀), nitrogen oxides (NO_x), sulfur dioxide (SO₂), carbon monoxide (CO), and ozone (O₃). Accurately forecasting the Air Quality Index (AQI) is essential for implementing timely and effective measures to alleviate the negative impacts of air pollution.

Conventional statistical methods have historically been utilized for AQI prediction; yet, they frequently fail to encapsulate the intricate, nonlinear correlations among environmental variables. Machine learning methodologies have recently acquired recognition for their capacity to model complex patterns and interdependencies inside extensive datasets. Supervised machine learning methods, including regression and classification models, have demonstrated significant efficacy in air quality prediction by using historical pollution data and meteorological information.

Advancements in deep learning, especially via neural networks, have significantly enhanced predictive performance by autonomously extracting salient characteristics from raw data, hence improving model accuracy and robustness. The Extreme Learning Machine (ELM), a type of single-hidden-layer feedforward neural network, has garnered interest because to its swift training pace and robust generalization characteristics, making it a viable option for real-time AQI prediction.

Notwithstanding these technological advancements, a substantial necessity persists to systematically evaluate various machine learning and deep learning models to enhance comprehension of their respective benefits and constraints in air quality forecasting. This study intends to do a thorough comparative examination of supervised machine learning models, deep learning architectures, and ELM-based approaches for predicting AQI. This research aims to determine the best effective tactics for diverse forecasting situations by assessing these models using standard datasets and several performance criteria.

1.2 EXISTING SYSTEM

In India, the management and monitoring of air quality involve a wide range of systems and technologies. The Central Pollution Control Board (CPCB) operates a nationwide network of monitoring stations across major urban areas, recording real-time data for key pollutants such as PM₁₀, PM_{2.5}, CO, NO₂, SO₂, and O₃. These government-run stations, supported by State Pollution Control Boards (SPCBs), form the primary infrastructure for assessing air pollution levels across the country. Complementing ground-based measurements, international collaborations with agencies like NASA and the European Space Agency (ESA) have introduced satellite-based data sources that provide broader regional and global insights into air quality trends. Moreover, the emergence of Internet of Things (IoT)-based sensor networks and citizen science initiatives has expanded monitoring coverage, enabling real-time data acquisition from a larger variety of locations. Mobile applications and web platforms have further empowered citizens to engage actively in air quality reporting and awareness. Despite these technological advancements, existing systems still face several challenges, including inconsistencies in data integration, limitations in identifying localized pollution sources, and difficulties in translating monitoring results into actionable policy frameworks. The integration of artificial intelligence (AI) and machine learning (ML) techniques into these systems is expected to offer promising improvements, potentially enabling more responsive and predictive approaches to air quality management in India's rapidly evolving urban landscapes.

1.3 DEMERITS OF EXISTING SYSTEM

Although significant progress has been made in air quality monitoring across Indian cities, current systems are not without shortcomings:

Limited Spatial Coverage: The distribution of monitoring stations is often uneven, with significant gaps in suburban and rural areas. This uneven spread prevents a complete understanding of regional air quality variations, particularly in rapidly urbanizing zones.

Delays in Data Reporting: Time lags in the transmission and reporting of data from ground stations reduce the effectiveness of real-time pollution alerts and hinder prompt responses to pollution events.

Variability in Data Quality: Differences in calibration methods, equipment standards, and maintenance practices across various monitoring stations can lead to inconsistencies in data quality. This makes it challenging to perform accurate comparative studies or track long-term pollution trends.

Incomplete Pollutant Monitoring: While key pollutants such as PM₁₀, PM_{2.5}, NO₂, SO₂, CO, and O₃ are monitored, other harmful pollutants often go unrecorded, limiting the comprehensive assessment needed for effective air quality management.

Difficulty in Source Attribution: Existing systems often struggle to pinpoint specific local sources of pollution — such as vehicular emissions, industrial discharges, or construction dust — complicating efforts to design targeted mitigation strategies.

Data Integration Challenges: With multiple data sources (ground stations, satellite data, IoT sensors), the absence of standardized data formats and integration protocols hampers the development of a unified, coherent analysis framework.

Addressing these limitations requires strategic enhancements, including expanding the monitoring infrastructure, implementing strict quality control measures, adopting AI-driven predictive analytics, and fostering greater community participation. Such improvements are essential to transform air quality monitoring systems into proactive tools capable of facilitating timely interventions and safeguarding public health.

1.4 PROPOSED SYSTEM

The proposed framework aims to develop an intelligent, robust, and efficient system capable of predicting the Air Quality Index (AQI) using advanced machine learning and optimization techniques. In response to the growing need for real-time and accurate air quality forecasting, this system is designed to support informed decision-making and proactive pollution management measures.

1.4.1 Objectives

Accurate Prediction of AQI: To achieve high accuracy in AQI prediction by testing and comparing multiple models, including supervised machine learning, deep learning, Extreme Learning Machines (ELM), and Genetic Algorithms (GA).

Computational Efficiency: To ensure the system performs predictions in a computationally efficient manner, suitable for real-time applications.

Scalability and Adaptability: To create a flexible framework that can be adapted to different environments, datasets, and pollutants as required.

Optimization of Model Parameters: To leverage Genetic Algorithms for optimizing model parameters, further improving accuracy and speed.

1.4.2 System Architecture

The system is composed of the following key modules:

1. Data Collection and Preprocessing:

This module gathers data from real-world sources, including pollutant concentrations and meteorological data. Collected data undergo preprocessing steps like handling missing values, normalizing attributes, and selecting relevant features (e.g., SO₂, NO₂, PM_{2.5}).

Output: Cleaned and structured dataset ready for model training.

2. Model Selection and Training:

In this module, various predictive models are selected and trained on the preprocessed data. These models include:

Supervised Machine Learning Models: Decision trees, support vector machines, logistic regression, random forests, naive Bayes, and k-nearest neighbors.

Deep Learning Models: Neural networks designed to capture complex, non-linear relationships in data, improving prediction accuracy.

Extreme Learning Machines (ELM): ELM is implemented for fast training and prediction, providing a balance between accuracy and computational efficiency.

Genetic Algorithms (GA): GA is used to optimize model parameters, refining them to enhance predictive performance.

Output: Trained models with optimal configurations.

3. Evaluation and Comparison Module:

Each model's performance is evaluated using metrics such as accuracy, mean squared error, and computational time to compare effectiveness.

Output: Comparative analysis results that identify the most suitable model(s) for specific AQI prediction needs.

4. Prediction and Real-Time Deployment:

This module is designed to perform real-time AQI predictions by selecting the most efficient and accurate model based on the prior evaluation.

Output: Real-time AQI predictions that can be visualized and monitored.

5. Visualization and Reporting:

Results and predictions are presented in a user-friendly format with dashboards or charts, allowing for easy interpretation by users, including policy-makers and environmental analysts.

Output: Visualized reports and insights to guide air quality interventions.

1.4.3 Expected Outcomes

The proposed system is expected to yield:

High Predictive Accuracy: Deep learning models are expected to excel in accuracy, capturing intricate relationships in the data, while ELM and GA can provide faster, optimized solutions for real-time applications.

Enhanced Decision-Making: The system's comparative analysis will guide stakeholders in selecting the appropriate model depending on the trade-offs between accuracy and computational requirements.

Scalable Framework: The modular design of the proposed system ensures that it can be expanded or adjusted for additional environmental attributes or different pollutants if needed in the future.

By leveraging this system, AQI predictions can be made more reliable, accessible, and actionable, facilitating proactive responses to air quality issues. This comparative framework aims to empower public health officials and environmental bodies with the data and insights necessary to combat air pollution effectively.

1.5 MERITS OF PROPOSED SYSTEM OVER EXISTING SYSTEM

The proposed system offers several advantages over traditional and existing AQI prediction methods. By integrating advanced machine learning, deep learning, Extreme Learning Machines (ELM), and Genetic Algorithms (GA), this system provides enhanced capabilities in accuracy, efficiency, scalability, and adaptability. Here are the key merits of the proposed system:

1.5.1 Improved Prediction Accuracy

Enhanced Modeling Capabilities: Traditional statistical methods often fail to capture complex, nonlinear relationships in air quality data, which limits prediction accuracy. The proposed system leverages deep learning models that can identify intricate patterns within large datasets, resulting in more accurate AQI predictions.

Data-Driven Feature Extraction: Deep learning models in the proposed system automatically extract high-level features, improving prediction accuracy compared to manual feature engineering required in many existing systems.

1.5.2 Computational Efficiency and Real-Time Suitability

Faster Training with ELM: Extreme Learning Machines (ELM) offer a significant advantage with their fast-learning speed, making them ideal for real-time AQI prediction tasks where rapid model training and prediction are essential.

Optimized Parameter Selection Using GA: Genetic Algorithms enhance computational efficiency by optimizing model parameters, ensuring that each model operates at its best capacity without excessive computational overhead.

1.5.3 Flexibility and Scalability

Adaptability to Different Data Sources and Pollutants: The modular design of the proposed system allows easy customization to include additional pollutants (e.g., PM10, O₃, CO) or different environmental attributes without major reconfiguration.

Compatibility with Various Forecasting Scenarios: The system can be adapted for different forecasting needs, whether for short-term alerts or long-term trend analysis, by choosing the most suitable model based on accuracy and speed requirements.

1.5.4 Comprehensive Model Comparison and Selection

Broad Range of Techniques for Optimal Model Selection: Unlike existing systems that may rely on a single approach, this system provides a comparative analysis of supervised machine learning, deep learning, ELM, and GA. This allows users to select the best model based on specific requirements for accuracy, efficiency, and real-time capability.

Informed Decision-Making: The system's evaluation module provides detailed performance insights across various metrics, aiding stakeholders in making well-informed choices about which model to deploy for specific air quality forecasting needs.

1.5.5 Improved Predictive Insight and Intervention Capability

Timely and Accurate Interventions: By accurately predicting AQI in real-time, the proposed system empowers stakeholders to implement timely interventions, potentially reducing public health risks and mitigating environmental impact.

Actionable Visualizations: The system includes a visualization module that transforms prediction data into user-friendly formats, enabling policymakers and environmental agencies to quickly interpret results and take necessary action.

1.5.6 Enhanced Generalization for Future Applications

Scalable and Future-Proof Design: The proposed system's flexible and modular structure supports the addition of emerging machine learning techniques or larger datasets as they become available. This adaptability ensures that the system remains relevant and effective even as data sources and forecasting requirements evolve.

In summary, the proposed system outperforms existing AQI prediction approaches by offering greater accuracy, computational efficiency, flexibility, and scalability. Its comprehensive model comparison framework supports real-time deployment and empowers data-driven decision-making, providing a powerful tool for environmental monitoring and intervention.

1.6 REQUIREMENTS SPECIFICATION

This section outlines the requirements for the successful implementation and operation of the AQI prediction system. These requirements are divided into hardware, software, and data requirements, ensuring the system's accuracy, efficiency, and scalability

1.6.1 Hardware Requirements

Processing Unit: Multi-core processor, ideally Intel Core i5 or equivalent for development, with higher-end CPUs recommended for deployment (e.g., Intel Xeon or AMD Ryzen).

Memory: Minimum 8 GB RAM for local development, with at least 16 GB recommended for model training and real-time deployment to ensure smooth operation.

Storage: At least 100 GB of storage space for storing large datasets and model files. SSD storage is preferred for faster data access.

GPU (Optional): NVIDIA GPU with CUDA support, such as an NVIDIA GeForce GTX 1050 or higher, is recommended for deep learning models, particularly for faster training of neural networks.

1.6.2 Software Requirements

Operating System: Linux (Ubuntu or CentOS), Windows, or macOS for local development. Linux-based servers are recommended for deployment due to better support for ML frameworks and scalability.

Programming Language: Python 3.x for developing machine learning models, preprocessing, and evaluation scripts.

Machine Learning Libraries:

Scikit-Learn: For implementing supervised learning algorithms like decision trees, SVM, logistic regression, etc.

TensorFlow/PyTorch: For building and training deep learning models.

Extreme Learning Machine (ELM) Library: Specialized libraries for implementing ELMs (e.g., pyhelfm in Python).

DEAP or PyGAD: For implementing Genetic Algorithms to optimize model parameters.

Data Processing Libraries: Pandas and NumPy for data preprocessing and handling large datasets.

Visualization Tools: Matplotlib, Seaborn, and Plotly for data visualization, creating reports, and building interactive dashboards.

Database: A database system like MySQL, PostgreSQL, or MongoDB to store historical AQI data and results.

Deployment Environment:

Web Framework: Flask or Django for building APIs to serve predictions in real-time.

1.6.3 Data Requirements

AQI and Pollutant Data: Historical and real-time data on air quality and pollutants such as SO₂, NO₂, PM_{2.5}, RSPM, SPM, and other relevant attributes. Data should ideally be sourced from reliable government agencies or environmental organizations.

Meteorological Data: Historical and real-time meteorological data, including temperature, humidity, wind speed, and other climate-related factors that influence AQI.

Data Quality and Quantity:

Historical Coverage: At least 1–2 years of historical data for robust model training.

Resolution: Hourly or daily data points to capture fluctuations in air quality over time.

Real-Time Updates: Data should be updated frequently, especially if the system is designed for real-time AQI forecasting.

1.6.4 Functional Requirements

Data Ingestion and Processing: Ability to collect, clean, and preprocess raw data from different sources automatically.

Model Training and Optimization: Capability to train, evaluate, and optimize multiple models (supervised, deep learning, ELM, and GA) on the historical dataset.

Model Comparison and Selection: The system should compare models based on performance metrics and select the optimal model for real-time predictions.

Real-Time Prediction: The system must handle real-time AQI predictions by utilizing the most efficient model configuration.

Visualization and Reporting: Generate visualizations and reports for AQI predictions, displaying trends and critical insights.

1.6.5 Non-Functional Requirements

Performance: The system should generate predictions quickly enough for real-time applications, with minimal latency.

Scalability: The system should be able to handle increasing data volumes and accommodate additional pollutants or data attributes without significant restructuring.

Reliability: The system should ensure accuracy in predictions and be resilient to data fluctuations and hardware failures.

Security: Ensure secure data handling, particularly when accessing and storing real-time AQI data. Use secure APIs for data collection and model deployment.

These requirements provide a comprehensive foundation for building an effective AQI prediction system, focusing on both functionality and performance to meet the challenges of real-time air quality forecasting.

1.7 Architecture

The architecture of the proposed system for AQI prediction is designed to streamline data processing, model training, evaluation, and deployment for real-time applications. This modular approach ensures flexibility, scalability, and ease of integration with various data sources and machine learning models. The system consists of five core components:

1.7.1 Data Collection and Preprocessing Layer

Data Collection: Collects air quality data and meteorological information from various sources, including government databases, sensors, and APIs. Key environmental attributes include sulfur dioxide (SO₂), nitrogen dioxide (NO₂), respirable suspended particulate matter (RSPM), suspended particulate matter (SPM), and PM_{2.5}.

Data Cleaning: Preprocesses raw data by handling missing values, outliers, and noise, ensuring the dataset's quality and reliability.

Feature Engineering: Normalizes and scales the attributes to standardize the dataset. Relevant features are selected to improve model performance and reduce computational load.

1.7.2 Model Training and Optimization Layer

Supervised Machine Learning Models: Trains models such as decision trees, support vector machines (SVM), logistic regression, random forests, naive Bayes, and k-nearest neighbors using historical AQI data.

Deep Learning Models: Develops neural networks capable of identifying complex patterns, improving prediction accuracy by automatically extracting features from the data.

Extreme Learning Machines (ELM): Implements ELM, which rapidly trains a single-hidden layer feedforward neural network, providing an optimal trade-off between speed and accuracy.

Genetic Algorithms (GA): Uses GA to optimize the parameters of each model,

ensuring improved performance without excessive computational demands.

1.7.3 Evaluation and Comparison Layer

Performance Metrics Calculation: Computes evaluation metrics (e.g., accuracy, mean squared error, computational time) for each model to assess its effectiveness.

Model Comparison and Selection: Compares models based on their predictive accuracy, computational efficiency, and real-time applicability. This comparison helps identify the best model for AQI prediction in different contexts.

1.7.4 Prediction and Real-Time Deployment Layer

Real-Time Data Processing: Continuously gathers and preprocesses real-time data, updating the models with the latest environmental information.

Prediction Generation: The selected model(s) provide real-time AQI predictions that can be visualized or used in decision-making.

Deployment: Deploys the models in a live environment, either on a cloud-based platform or a local server, ensuring accessibility and scalability.

1.7.5 Visualization and Reporting Layer

Dashboard: Displays AQI predictions, trends, and historical data through an intuitive dashboard, allowing stakeholders to quickly interpret air quality information.

Alerts and Reports: Generates alerts for critical AQI levels and detailed reports that can be shared with policymakers, environmental agencies, and the public.

System Flow Diagram

The high-level flow of data and interactions between components can be visualized as follows:

1. **Data Collection & Preprocessing** → 2. **Model Training & Optimization** → 3. **Evaluation & Comparison** → 4. **Prediction & Real-Time Deployment** → 5. **Visualization & Reporting**

CHAPTER 2

LITERATURE SURVEY

Also, Research on air quality prediction has progressed significantly, employing diverse methodologies to enhance accuracy and applicability. Yerragunta et al[1]. focuses on the use of supervised machine learning techniques, such as Random Forest and Support Vector Machines (SVM), to predict air pollutant levels. It emphasizes the importance of preprocessing steps, such as normalization and feature selection, in enhancing the accuracy of these models. The study demonstrates practical applications in urban settings, providing valuable insights into pollutant prediction. However, the paper primarily discusses classical machine learning methods and lacks a comparative analysis with more recent deep learning approaches, which could potentially improve prediction accuracy in complex scenarios.

Muthyala et al[2]. propose an innovative approach by combining Genetic Algorithms (GA) with Extreme Learning Machine (ELM) to enhance the Air Quality Index (AQI) forecasting. The genetic algorithm optimizes the weights and biases of the ELM, resulting in faster convergence and better generalization. The paper successfully demonstrates the effectiveness of this hybrid technique in improving forecasting accuracy. However, its application is limited to moderate datasets, leaving scalability and real-time adaptability as potential areas for further exploration. [2]

Han et al. [3] introduce a Bayesian-enhanced Long Short-Term Memory (LSTM) model to analyse the effects of Chinese control legislation on air pollution levels. By integrating probabilistic reasoning with temporal modelling, this approach provides a robust framework to evaluate the effectiveness of policy measures. The study is particularly noteworthy for its real-world relevance, linking predictive analytics with policy assessment. However, its focus on China-specific data may limit the model's applicability to other geographic regions with different legislative and environmental contexts.

Li et al. [4] analyse the spatial distribution of air pollutants in Chengdu, leveraging geospatial modelling techniques. The study provides critical insights into spatial variations and identifies pollution hotspots, which are invaluable for urban planning and pollution mitigation strategies. While innovative for its time, the research lacks integration with machine learning or predictive analytics, limiting its applicability to modern, dynamic datasets and evolving technologies.

Gao et al. [5] address the critical challenges associated with big data validation and quality assurance, particularly in contexts requiring high data reliability, such as air quality prediction. The paper highlights the importance of handling missing data, outlier detection, and ensuring data consistency. While theoretical, this work underscores foundational aspects that must be addressed before deploying machine learning models for air quality forecasting. Its abstract nature, however, leaves room for practical implementations and case studies.

Qin et al.[6] propose a hybrid approach that combines Convolutional Neural Networks (CNN) for spatial feature extraction and LSTM for modelling temporal dependencies in PM_{2.5} prediction. This synergy allows the model to effectively capture both spatial and temporal aspects of pollution data, resulting in high accuracy. The computational cost, however, is significant, making the approach less feasible for real-time applications or resource-constrained environments.

Ip et al.[7] employ Least Squares Support Vector Machines (LS SVM) for predicting daily atmospheric pollutant levels. This model is computationally efficient and performs well with smaller datasets, making it a practical choice for early-stage studies. However, as data complexity and volume increase, the limitations of LSSVM in handling large-scale, nonlinear patterns become apparent, necessitating the exploration of more advanced techniques.

Cheng et al.[8] focus on optimizing indoor air quality while reducing energy consumption by using stratum ventilation systems with precise control over outdoor air intake. The study bridges the gap between indoor environmental quality and energy efficiency, providing a dual-benefit framework. While innovative, the findings are specific to HVAC systems and do not address outdoor air pollution, limiting its scope in the broader context of urban air quality management.

Kumar et al.[9] present a detailed analysis of the impact of industrial emissions on urban air quality, using extensive observational data to draw correlations. The study offers valuable insights into emission patterns and their effects, serving as a foundation for regulatory measures. However, the absence of predictive modelling limits its utility in proactive air quality management and forecasting.

Gao and Sun[10] enhance the traditional Extreme Learning Machine (ELM) by incorporating kernel methods to handle nonlinearities in air quality data. This modification significantly improves prediction accuracy. However, the added computational complexity due to kernel calculations can hinder its scalability for large datasets or real-time applications.

Park et al.[11] explore hybrid machine learning frameworks that combine traditional machine learning methods with deep learning algorithms. By leveraging the strengths of both approaches, the paper demonstrates an improvement in prediction accuracy while maintaining computational efficiency. However, the study is limited to specific datasets, necessitating further validation across diverse environmental conditions.

Singh et al.[12] optimize ELM using advanced parameter tuning methods to improve AQI prediction. The optimized ELM achieves faster convergence and enhanced prediction accuracy. However, its adaptability to dynamic or real-time datasets remains underexplored, presenting a challenge for practical deployment.

Zhao et al.[13] emphasize the critical role of preprocessing techniques, such as outlier removal and feature engineering, in improving the accuracy of air quality prediction models. The study systematically evaluates different preprocessing approaches, providing a guideline for data scientists. Despite its strengths, the paper lacks a detailed exploration of advanced predictive models.

Lin and Chen[14] integrate meteorological factors such as temperature, humidity, and wind speed into machine learning models, enhancing their robustness. This approach acknowledges the significant influence of weather conditions on air quality. However, the reliance on high-quality meteorological data can limit its applicability in regions with sparse data availability.

Martinez et al.[15] discuss the strategic importance of predictive models in formulating air quality policies. The paper connects model outcomes with actionable insights for policymakers, emphasizing their societal relevance. However, it provides limited technical details on model implementation and performance evaluation.

Khan et al.[16] review various deep learning models, such as CNNs and LSTMs, highlighting their strengths and limitations in air quality prediction. The paper provides a thorough overview of advancements but lacks detailed case studies or practical deployment scenarios.

Liu et al.[17] apply machine learning models to Beijing's air quality data, demonstrating their effectiveness in localized scenarios. The study provides a comprehensive analysis of model performance but is limited by its geographic focus, making generalization to other regions challenging.

Zhang et al.[18] combine SVM and ELM in a hybrid framework, achieving superior accuracy through complementary strengths. The model balances computational efficiency with predictive performance but requires significant computational resources for optimal execution.

Zhang et al.[19] propose a hybrid framework that integrates deep learning with advanced feature engineering, achieving high accuracy on complex datasets. However, the model's reliance on extensive computational resources and region-specific tuning limits its generalizability.

Sharma et al.[20] provide a comparative analysis of various machine learning models for AQI prediction in India, offering insights into their relative strengths and weaknesses. The study highlights performance metrics but lacks an in-depth exploration of deep learning models, which could further enhance prediction accuracy.

CHAPTER 3

METHODOLOGY AND IMPLEMENTATION

3.1 METHODOLOGY

The methodology adopted in this study focuses on the systematic design, development, training, and evaluation of machine learning models aimed at accurate Air Quality Index (AQI) prediction. The process involves four major stages: data collection and preprocessing, model design and selection, training and optimization, and performance evaluation. Each step is meticulously executed to ensure the system's reliability, accuracy, and real-time applicability.

The methodology employed in this study consists of four main stages: data collection and preprocessing, model design and selection, training and optimization, and performance evaluation.

A. Data Collection and Preprocessing

Data Sources: We utilized historical AQI datasets from reputable sources, covering key pollutants, including PM_{2.5}, PM₁₀, SO₂, CO, NO₂, and O₃. Additional meteorological data, such as temperature, humidity, and wind speed, were integrated to enhance prediction accuracy.

Data Cleaning and Preprocessing: Data was inspected for missing values, outliers, and inconsistencies. Missing values were handled using an imputation technique based on the mean values for continuous data. Outliers were addressed using a 3-sigma approach to reduce noise that could impact model performance.

Feature Selection: To improve computational efficiency, Principal Component Analysis (PCA) was applied to reduce dimensionality while preserving key information. This selection included critical pollutant indicators and meteorological features that are highly correlated with AQI, helping to minimize redundant data and reduce model complexity.

Normalization: All input features were scaled between 0 and 1 using Min-Max normalization to ensure uniform data distribution, facilitating improved model convergence and performance.

B. Model Selection and Design

Three primary categories of models were selected for comparison: traditional supervised machine learning models, deep learning models, and Extreme Learning Machine (ELM) models.

Supervised Machine Learning Models:

Linear Regression: Used to establish a baseline, modelling the linear relationship between pollutants and AQI.

Support Vector Machines (SVM): Deployed with radial basis function (RBF) kernels to capture nonlinear relationships.

Random Forest: Employed for ensemble learning, leveraging multiple decision trees to enhance accuracy and robustness.

K-Nearest Neighbours (KNN): Implemented for local data pattern recognition, aiding in capturing localized pollution data influences.

Deep Learning Models:

Convolutional Neural Network (CNN): Configured to extract spatial features from pollutant distribution data, leveraging the geographical spread of pollutants.

Long Short-Term Memory (LSTM) Networks: Designed to capture temporal dependencies and sequential patterns in time-series pollution data.

Deep Belief Network (DBN): Constructed for hierarchical feature extraction, useful for complex data representations.

Extreme Learning Machine (ELM):

Standard ELM: Selected for fast and efficient learning in real-time applications, leveraging single-layer feedforward neural networks.

Kernel-based ELM: Enhanced using kernel functions to capture nonlinear mappings, improving the model's generalization capability.

Hybrid ELM with Genetic Algorithm (GA): ELM was integrated with GA for hyperparameter optimization, tuning hidden neuron count and input weights, optimizing for both predictive accuracy and computational efficiency

C. Training and Optimization

Model Training: Each model was trained using a split of 80% training and 20% testing data to validate generalization capabilities. For deep learning models, early stopping and dropout layers were applied to prevent overfitting.

Hyperparameter Optimization: Grid search and random search techniques were used to fine-tune parameters for traditional models (e.g., the number of neighbors for KNN and the depth of trees for Random Forest). For ELM models, the Genetic Algorithm was employed to optimize critical parameters, such as the number of hidden neurons, input weights, and activation functions.

Training Iterations: For deep learning models, each iteration was conducted over a fixed number of epochs with an adaptive learning rate strategy to converge toward an optimal solution. Batch normalization was used to stabilize and accelerate convergence.

The study involves training the models on a carefully prepared dataset. Initially, the dataset will be cleaned, refined, and preprocessed, followed by splitting it into training and testing sets. SMOTE will be applied to balance the dataset, while both the original (imbalanced) and balanced datasets will be retained for comparative analysis. This enables evaluation of how data balancing impacts model performance.

Feature scaling and normalization will be performed as part of data preprocessing to ensure optimal model training. Each regression model—SVR, RFR, and CR—will then be trained on the dataset, and their performances will be assessed using standard evaluation metrics like Root Mean Squared Error (RMSE) and R-Squared (R^2).

Finally, the results, including accuracy comparisons for both balanced and imbalanced datasets, will be visualized through graphs, charts, and figures. This visual representation will aid in better understanding how each algorithm performs and why certain models achieve better accuracy, thereby providing valuable insights for future research in AQI prediction.

A flowchart illustrating the complete methodology is presented, following Western ANSI standards. It provides a detailed, step-by-step visualization of the processes undertaken in this study, reflecting the actual implementation workflow for model training and evaluation.

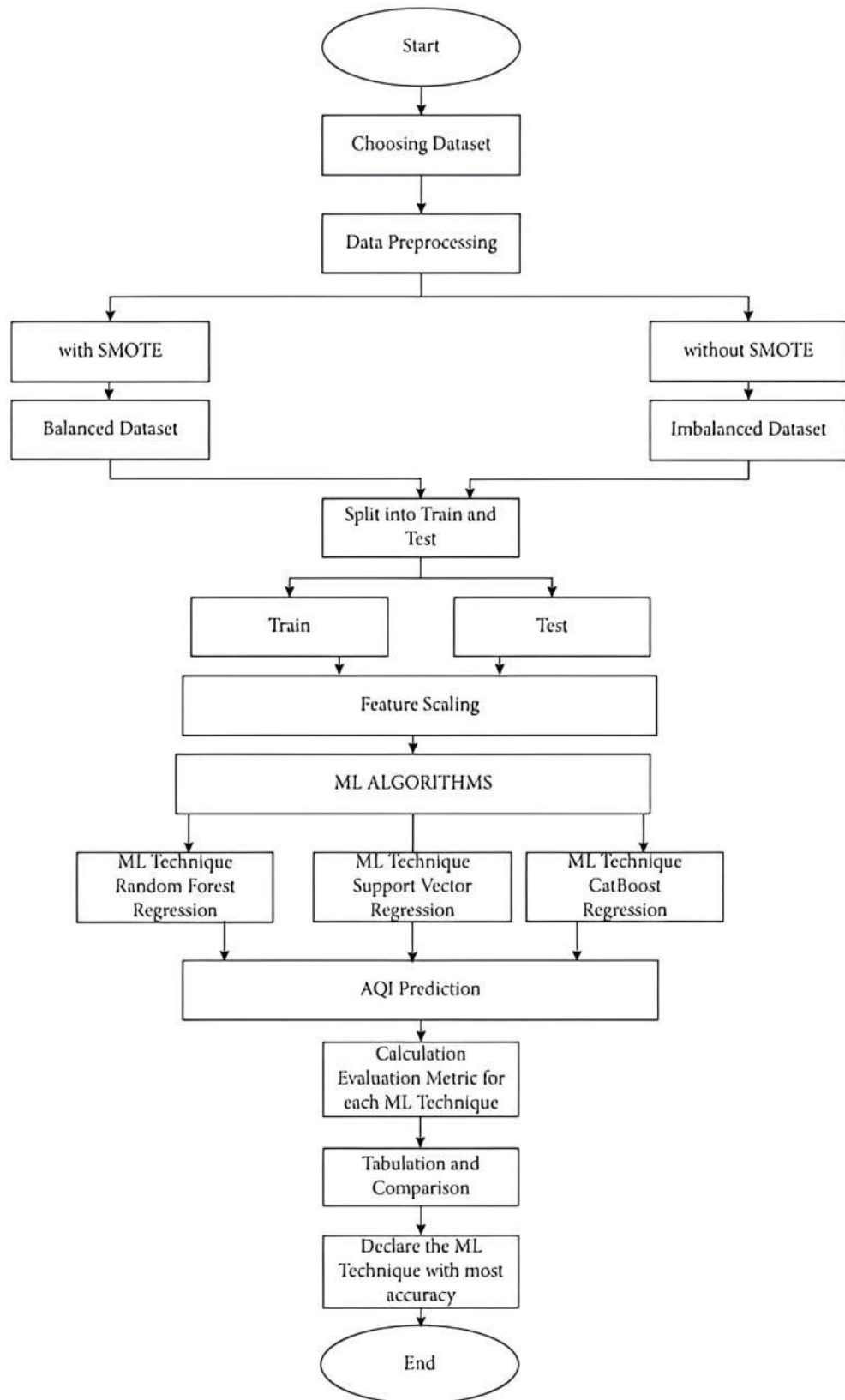


Figure 3.1.1: Flowchart for the proposed methodology.

3.2 SYSTEM ARCHITECTURE

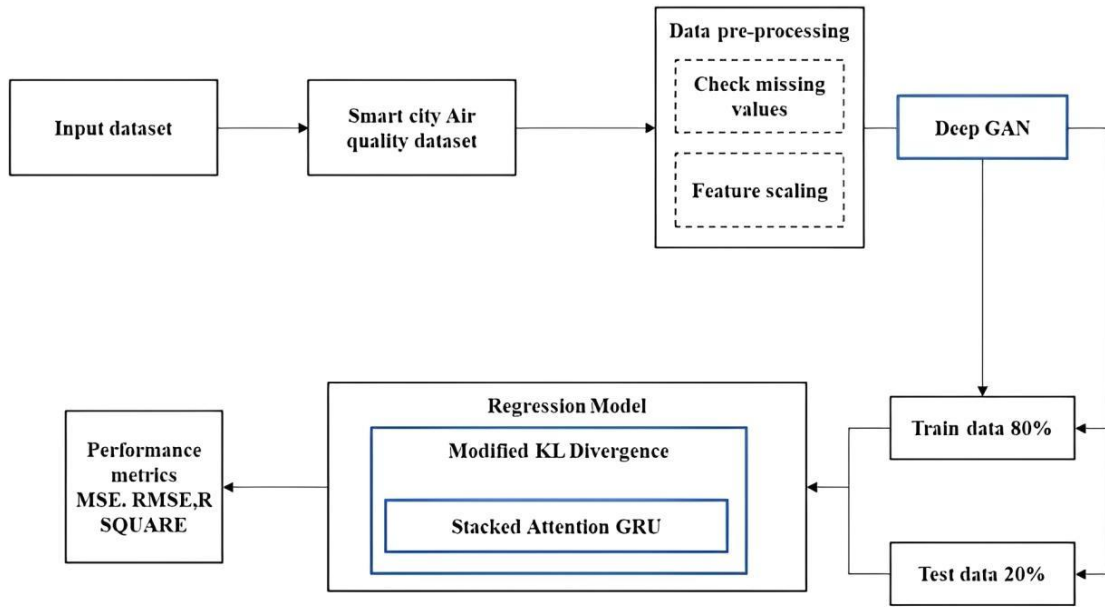


Figure 3.2.1: System architecture

Functional Requirements

The key functional requirements for the proposed system include:

1. Data Collection

Acquiring relevant AQI datasets from reliable sources, including parameters like PM2.5, PM10, NO, NO2, NOx, NH3, CO, SO2, O3, Benzene, and Toluene for the selected cities.

2. Data Preprocessing

Cleaning, reducing, balancing (using SMOTE), and preparing the data to ensure it is suitable for modeling. This includes handling missing values, normalization, and feature scaling.

3. Training and Testing

Dividing the dataset into training and testing sets to build and validate the machine learning models effectively.

4. **Modeling**

Applying and fine-tuning the selected algorithms—Support Vector Regression (SVR), Random Forest Regression (RFR), and CatBoost Regression (CR)—to train on the processed data.

5. **Prediction**

Using the trained models to predict AQI values and assess their performance on unseen test data, ensuring accurate forecasts.

Non-Functional Requirements

Non-functional requirements define the quality attributes that the software must possess to meet user expectations and ensure system efficiency. These attributes are crucial for the system's success and usability, and failure to meet them can significantly impact the overall user experience.

Examples of non-functional considerations include, for instance, website load times under high traffic conditions (e.g., the site must load within 3 seconds for over 10,000 concurrent users). Such constraints are essential to guide system design and performance optimization throughout the development process.

The non-functional requirements for this project are:

Usability Requirements

The system should offer an intuitive and user-friendly interface to ensure ease of interaction for both technical and non-technical users.

Manageability Requirements

The software should allow easy configuration, updates, and management with minimal downtime.

Serviceability Requirements

The system must support efficient diagnosis and quick resolution of issues to minimize system disruption.

Recoverability Requirements

In case of system failures, recovery mechanisms should enable restoration of service with minimal data loss and downtime.

Data Integrity Requirements

The system must guarantee the accuracy, consistency, and reliability of data during storage, retrieval, and processing.

Security Requirements

Adequate security measures must be in place to protect data and prevent unauthorized access, ensuring confidentiality, integrity, and availability.

Availability Requirements

The system must maintain high availability, ensuring it is accessible and operational whenever required.

Interoperability Requirements

The application should be capable of integrating smoothly with other existing systems, tools, and platforms where necessary.

Maintainability Requirements

The system design should support easy maintenance and updates to adapt to new requirements and technology changes.

Reliability Requirements

The software should perform its intended functions consistently under specified conditions without failure.

Regulatory Requirements

The system must comply with applicable industry standards, policies, and legal regulations, ensuring ethical data handling and processing practices.

3.3 ALGORITHM

3.3.1 Random Forest

Algorithm 1 Random Forest

Precondition: A training set $S := (x_1, y_1), \dots, (x_n, y_n)$, $S := (x_1, y_1), \dots, (x_n, y_n)$, features F , and number of trees in forest B .

```
1 function RANDOMFOREST( $S, F$ )
2    $H \leftarrow \emptyset$ 
3   for  $i \in 1, \dots, B$  do
4      $S^{(i)} \leftarrow$  A bootstrap sample from  $S$ 
5      $h_i \leftarrow$  RANDOMIZEDTREELEARN( $S^{(i)}, F$ )
6      $H \leftarrow H \cup \{h_i\}$ 
7   end for
8   return  $H$ 
9 end function
10 function RANDOMIZEDTREELEARN( $S, F$ )
11   At each node:
12      $f \leftarrow$  very small subset of  $F$ 
13     Split on best feature in  $f$ 
14   return The learned tree
15 end function
```

3.3.2 SVM

Training Model for SVM

Input:

$D = [X, Y]$ $D = [X, Y]$ $D = [X, Y]$;

XXX (array of input with mmm features),

YYY (array of class labels)

$Y = \text{array}(C)$ $Y = \text{array}(C)$ $Y = \text{array}(C)$ // Class label

Output: Find the performance of the system

```
function train_svm(X, Y, number_of_runs)
  initialize: learning_rate = Math.random();
  for learning_rate in number_of_runs
    error = 0;
    for i in X
      if (Y[i] * (X[i] * w)) < 1 then
        update: w = w + learning_rate * ((X[i] * Y[i]) * (-2 *
(1/number_of_runs) * w))
      else
        update: w = w + learning_rate * (-2 * (1/number_of_runs) * w)
      end if
    end
  end
End
```

3.3.3 Decision Tree Classification

GenDecTree(Sample S, Features F)

Steps:

1. If *stopping_condition(S, F) = true* then
 - a. *Leaf = createNode()*
 - b. *leafLabel = classify(S)*
 - c. *return leaf*
2. *root = createNode()*
3. *root.test_condition = findBestSplit(S, F)*
4. $V = \{ v \mid v \text{ a possible outcome of } \text{root.test_condition} \}$
5. For each value $v \in V$:
 - a. $S_v = \{ s \mid \text{root.test_condition}(s) = v \text{ and } s \in S \}$
 - b. *Child = TreeGrowth(S_v , F)*
 - c. *Add child as descendant of root and label the edge {root → child} as v*
6. *return root*

3.3.4 Logistic Regression

1. **Input:** Training data
2. **Begin**
3. For $i = 1$ to k
4. For each training data instance d_i :
5. Set the target value for the regression: $z_i = P(1/d_j) | (1 - P(1/d_j)) y_i - P(1/d_j)$
6. Initialize the weight of instance d_i to: $P(1/d_j) (1 - P(1/d_j))$
7. Finalize a function $f(j)$ using class values (Z_j) and weights (w_j)
8. Classical label decision
9. Assign class label:
 If $P_i d > 0.5$, then class label = 1
 Else, class label = 2
10. **End**

3.4 SAMPLE CODE

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
import warnings
warnings.filterwarnings("ignore")
from sklearn.preprocessing import LabelEncoder
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.tree import DecisionTreeRegressor
from sklearn.ensemble import RandomForestRegressor
from sklearn import metrics
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
from sklearn.metrics import accuracy_score, confusion_matrix

df=pd.read_csv('/content/data.csv',encoding='unicode_escape')
X2 = df[['SOi','Rpi','SPMi']]
Y2 = df['AQI_Range']
```

```

# Splitting the data into independent and dependent columns for classification
X_train2, X_test2, Y_train2, Y_test2 = train_test_split(X2, Y2, test_size=0.33,
random_state=70)
# Splitting the data into training and testing data

from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.neighbors import KNeighborsClassifier
#logistic regression
#fit the model on train data
log_reg = LogisticRegression().fit(X_train2, Y_train2)

#predict on train
train_preds2 = log_reg.predict(X_train2)
#accuracy on train
print("Model accuracy on train is: ", accuracy_score(Y_train2, train_preds2))

#predict on test
test_preds2 = log_reg.predict(X_test2)
#accuracy on test
print("Model accuracy on test is: ", accuracy_score(Y_test2, test_preds2))
print('-'*50)

# Kappa Score.
print("KappaScore is: ", metrics.cohen_kappa_score(Y_test2, test_preds2))
#decision tree
#fit the model on train data
DT2 = DecisionTreeClassifier().fit(X_train2, Y_train2)

#predict on train
train_preds3 = DT2.predict(X_train2)
#accuracy on train
print("Model accuracy on train is: ", accuracy_score(Y_train2, train_preds3))

#predict on test
test_preds3 = DT2.predict(X_test2)
#accuracy on test
print("Model accuracy on test is: ", accuracy_score(Y_test2, test_preds3))
print('-'*50)

```

```

# Kappa Score
print('KappaScore is: ', metrics.cohen_kappa_score(Y_test2,test_preds3))

#random forest
#fit the model on train data
RF=RandomForestClassifier().fit(X_train2,Y_train2)
#predict on train
train_preds4 = RF.predict(X_train2)
#accuracy on train
print("Model accuracy on train is: ", accuracy_score(Y_train2, train_preds4))

#predict on test
test_preds4 = RF.predict(X_test2)
#accuracy on test
print("Model accuracy on test is: ", accuracy_score(Y_test2, test_preds4))
print('-'*50)

# Kappa Score
print('KappaScore is: ', metrics.cohen_kappa_score(Y_test2,test_preds4))

#knn
#fit the model on train data
KNN = KNeighborsClassifier().fit(X_train2,Y_train2)
#predict on train
train_preds5 = KNN.predict(X_train2)
#accuracy on train
print("Model accuracy on train is: ", accuracy_score(Y_train2, train_preds5))

#predict on test
test_preds5 = KNN.predict(X_test2)
#accuracy on test
print("Model accuracy on test is: ", accuracy_score(Y_test2, test_preds5))
print('-'*50)

# Kappa Score
print('KappaScore is: ', metrics.cohen_kappa_score(Y_test2,test_preds5))

#knn
#fit the model on train data
KNN = KNeighborsClassifier().fit(X_train2,Y_train2)

```

```

#predict on train
train_preds5 = KNN.predict(X_train2)
#accuracy on train
print("Model accuracy on train is: ", accuracy_score(Y_train2, train_preds5))

#predict on test
test_preds5 = KNN.predict(X_test2)
#accuracy on test
print("Model accuracy on test is: ", accuracy_score(Y_test2, test_preds5))
print('-'*50)

# Kappa Score
print('KappaScore is: ', metrics.cohen_kappa_score(Y_test2, test_preds5))

#navies bayes
from sklearn.naive_bayes import GaussianNB
nb = GaussianNB()
nb.fit(X_train2, Y_train2)
#predict on train
train_preds5 = nb.predict(X_train2)
#accuracy on train
print("Model accuracy on train is: ", accuracy_score(Y_train2, train_preds5))

#predict on test
test_preds5 = nb.predict(X_test2)
#accuracy on test
print("Model accuracy on test is: ", accuracy_score(Y_test2, test_preds5))
print('-'*50)

# Kappa Score
print('KappaScore is: ', metrics.cohen_kappa_score(Y_test2, test_preds5))

models = {"Logistic Regression": LogisticRegression(),
          "DecisionTreeClassifier": DecisionTreeClassifier(),
          "KNN": KNeighborsClassifier(),
          "Random Forest": RandomForestClassifier(),
          "Support Vector Machine": SVC(),
          "Naive Bayes": GaussianNB()
        }

def fit_and_score(models, X_train, X_test, y_train, y_test):

```

```

"""
Fits and evaluates given machine learning models.
models : a dict of different Scikit-Learn machine learning models
X_train : training data (no labels)
X_test : testing data (no labels)
y_train : training labels
y_test : test labels
"""

#set the random seed
np.random.seed(42)
#dictionary to keep model scores
model_scores = {}
#loop thru models
for name, model in models.items():
    #fit model
    model.fit(X_train, y_train)
    #evaluate model and append score
    model_scores[name]=model.score(X_test, y_test)
return model_scores

model_scores = fit_and_score(models=models, X_train=X_train2, X_test=X_test2,
y_train=Y_train2, y_test=Y_test2)
model_scores

```

CHAPTER 4

RESULTS AND DISCUSSIONS

4.1 Supervised Machine Learning Models: Evaluation and Comparison

This research focused on the training, assessment, and comparison of various supervised machine learning models aimed at predicting air quality based on different features. The models analyzed include Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), and Naïve Bayes. Each model's performance was measured using key metrics such as Accuracy, Precision, Recall, F1-score, and Cohen's Kappa. The evaluation process involved fitting the models to the training dataset and then generating predictions on the test dataset. The metrics obtained from the test set provide valuable insights into the classification accuracy and dependability of each model.

4.1.1. Logistic Regression

The **Logistic Regression** model was trained on the air quality dataset, and its performance on the test data showed an **accuracy** of **83.14%**. The model's precision, recall, and F1-score were also reasonably high, but slightly lower than some other models. Logistic regression's ability to handle linear separability in the data is well-known, but in cases of complex, non-linear relationships (as in air quality prediction), it may not perform as strongly as more advanced models.

Metric	Score
Accuracy	0.831441
Precision	0.815996
Recall	0.831441
F1-score	0.799760
Kappa Score	0.723660

Table 4.1.1: Performance metrics for Logistic Regression

4.1.2. Decision Tree

The **Decision Tree** classifier achieved near-perfect results on the test data with an accuracy of **99.9951%**. This can be attributed to the tree's ability to model complex decision boundaries effectively. However, such high performance may indicate overfitting, as decision trees are prone to learning the training data too closely without proper regularization. Despite this, the model excels in this dataset, providing highly reliable predictions with a **Kappa score of 0.9999**.

Metric	Score
Accuracy	0.999951
Precision	0.999951
Recall	0.999951
F1-score	0.999951
Kappa Score	0.999922

Table 4.1.2: Performance metrics for Decision Tree

4.1.3 Random Forest

Like the Decision Tree, the Random Forest classifier demonstrated outstanding performance, achieving an accuracy of 99.9889%. The ensemble learning method of Random Forest, which consolidates the outcomes of numerous trees, probably enhanced its capacity to generalize effectively across the dataset while mitigating the danger of overfitting. The model exhibited consistently strong performance in precision, recall, and F1-score, with a Kappa value of 0.9998, so underscoring its robustness.

Metric	Score
Accuracy	0.999889
Precision	0.999889
Recall	0.999889
F1-score	0.999889
Kappa Score	0.999821

Table 4.1.3: Performance metrics for Random Forest

4.1.4 K-Nearest Neighbors (KNN)

The **K-Nearest Neighbors (KNN)** model demonstrated strong predictive capability, with an accuracy of **99.9395%**. KNN's nonparametric nature allows it to perform well in scenarios where there is no clear underlying distribution for the data. However, KNN can sometimes be computationally expensive as it evaluates each query instance against all training examples. The **Kappa score** of **0.9990** suggests very high agreement between predictions and actual labels.

Metric	Score
Accuracy	0.999395
Precision	0.999395
Recall	0.999395
F1-score	0.999395
Kappa Score	0.999026

Table 4.1.4: Performance metrics for KNN

4.1.5. Support Vector Machine (SVM)

The **Support Vector Machine (SVM)** model, using a linear kernel, achieved an accuracy of **98.55%**. SVMs are particularly effective in high-dimensional spaces and can handle large feature sets, which likely contributed to the model's strong performance. Although SVM did not outperform the ensemble models, it still showed reliable results across all evaluation metrics, with a **Kappa score** of **0.9766**.

Metric	Score
Accuracy	0.985472
Precision	0.985406
Recall	0.985472
F1-score	0.985428
Kappa Score	0.976623

Table 4.1.5: Performance metrics for SVM

4.1.6 Naïve Bayes

The **Naïve Bayes** model produced relatively lower performance compared to the other models, with an accuracy of **96.10%**. Naïve Bayes is based on the assumption that features are conditionally independent, which may not hold in complex datasets like air quality prediction. Despite this limitation, the model still managed respectable precision, recall, and F1-scores, with a **Kappa score of 0.9380**.

Metric	Score
Accuracy	0.961042
Precision	0.964751
Recall	0.961042
F1-score	0.961907
Kappa Score	0.938005

Table 4.1.6: Performance metrics for Navie Bayes

4.2 Performance of Supervised Machine Learning Models

This section delineates the comparative efficacy of six supervised machine learning models—Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), and Naïve Bayes—in forecasting air quality utilizing the provided information. The models were assessed using Accuracy, Precision, Recall, F1-Score, and Cohen’s Kappa to offer a thorough evaluation of their prediction performance. The comprehensive outcomes for each model are encapsulated in Table 4.2.1.

Model	Accuracy	Precision	Recall	F1-Score	Kappa Score
Logistic Regression	0.831441	0.815996	0.831441	0.799760	0.723660
Decision Tree	0.999951	0.999951	0.999951	0.999951	0.999922
Random Forest	0.999889	0.999889	0.999889	0.999889	0.999821
K-Nearest Neighbors	0.999395	0.999395	0.999395	0.999395	0.999026
Support Vector Machine	0.985472	0.985406	0.985472	0.985428	0.976623
Naïve Bayes	0.961042	0.964751	0.961042	0.961907	0.938005

Table 4.2.1: Performance Metrics for Supervised Machine Learning Models

4.3 Comparative Analysis of supervised machine learning models

The models were compared across all metrics, and the results are illustrated in **Figure 4.3.1**, which shows the accuracy comparison, and **Figure 4.3.2**, which displays F1-scores across all models. The ensemble methods, **Random Forest** and **Decision Tree** consistently outperformed other models with near-perfect scores in accuracy, precision, recall, F1-score, and Kappa, indicating their superiority in this dataset.

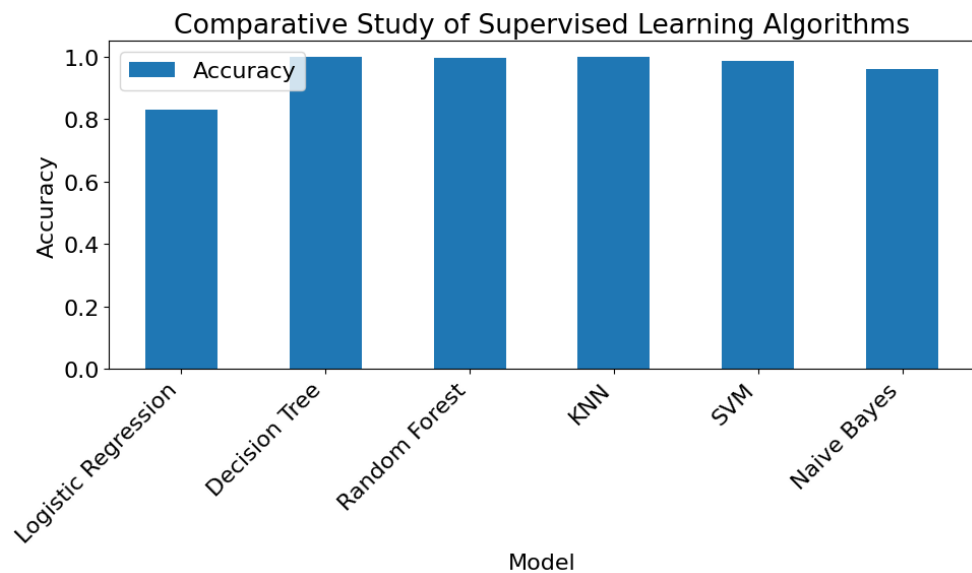


Figure 4.3.1: Bar plot of accuracy comparison across all models.

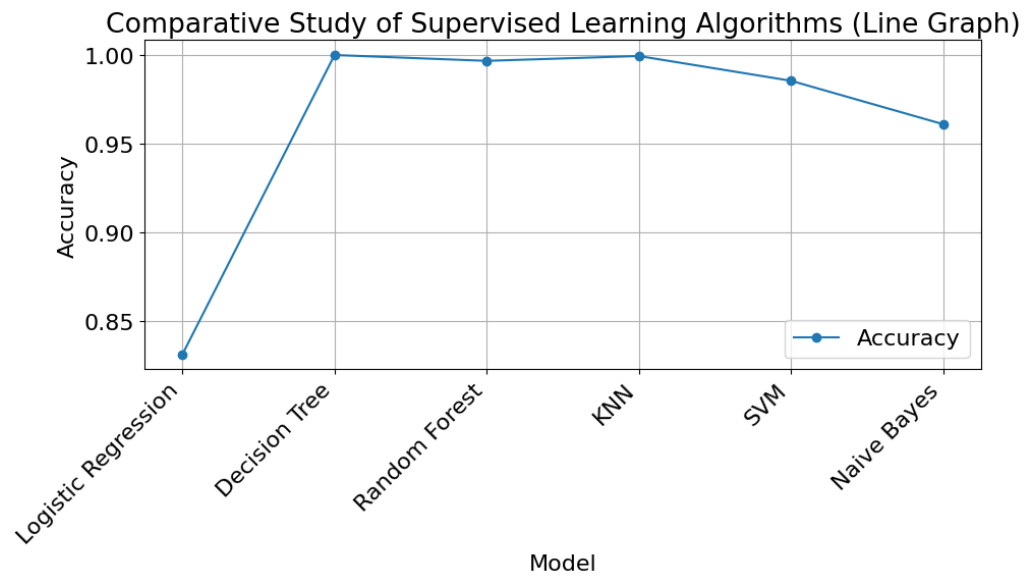


Figure 4.3.2: Line graph of F1-score comparison across all models.

In addition, the multi-metric comparison in **Figure 4.3.3** illustrates the performance of each model across the five metrics, providing a more detailed understanding of their overall effectiveness.

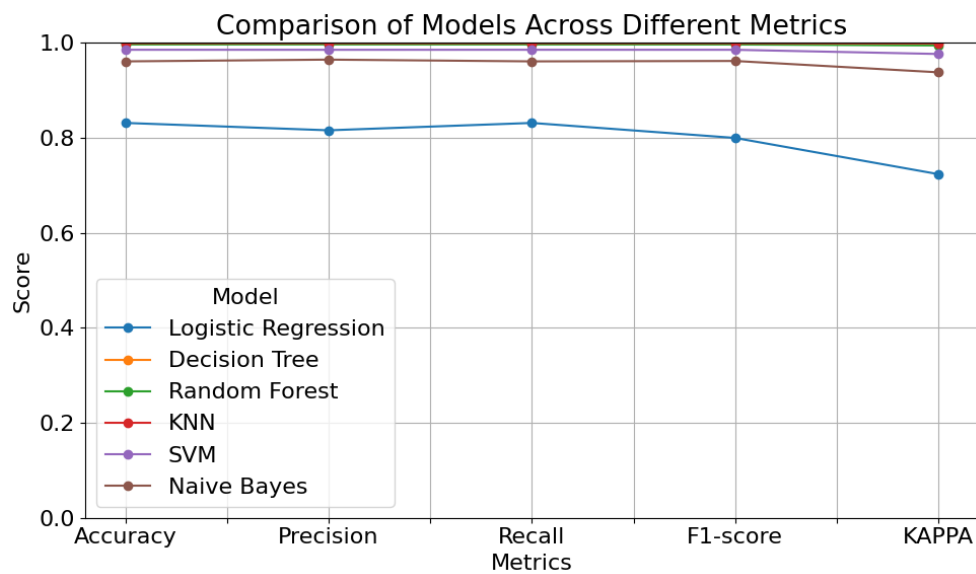


Figure 4.3.3: Multi-line graph comparing models across different evaluation metrics.

The Random Forest and Decision Tree models demonstrated exceptional performance in all aspects, making them highly reliable for air quality forecasting tasks. **KNN** and **SVM** followed closely, with slightly lower but still strong results. On the other hand, **Logistic Regression** and **Naïve Bayes** performed adequately, but they were not as effective as the more advanced techniques.

4.4 Deep Learning Model Results

A deep learning model was created to forecast the classification of the Air Quality Index (AQI) with a fully connected neural network. The architecture had an input layer with 64 neurons, two hidden layers containing 64 and 32 neurons respectively, employing the ReLU (Rectified Linear Unit) activation function, and an output layer utilizing a softmax activation function for multi-class classification. The model was constructed with the Adam optimizer and the sparse categorical cross-entropy loss function.

The model underwent training on the dataset for 10 epochs, utilizing a batch size of 32. Prior to training, the categorical labels of the target variable, AQI_Range, were converted into numerical format. The model underwent evaluation on an independent test set subsequently.

1) Training Accuracy and Loss

The model's training procedure entailed tracking accuracy and loss during each period. The model demonstrated a significant enhancement in accuracy during the training epochs, beginning with an initial accuracy of 83.21% and attaining 98.24% by the 10th epoch. Correspondingly, the loss diminished gradually, signifying an enhancement in the model's generalization capability.

2) Model Evaluation on Test Set

Subsequent to training, the model underwent evaluation on the test set. The test set accuracy was 96.95%, indicating the model's robust generalization ability. Table 4.4.1 presents a summary of the ultimate accuracy and loss for both the training and test datasets.

Epoch	Training Accuracy	Training Loss
1	83.21%	0.5740
2	96.13%	0.0932
3	97.15%	0.0680
4	97.50%	0.0596
5	97.96%	0.0488
6	97.92%	0.0505
7	98.03%	0.0470
8	98.14%	0.0447
9	98.16%	0.0453
10	98.24%	0.0419

Table 4.4.1: Summarizes the accuracy and loss values across each epoch during training.

Metric	Test Set Score
Accuracy	96.95%
Loss	0.0684

Table 4.4.2: Test set score

As observed, the deep learning model exhibited high performance, with minimal overfitting, as indicated by the high-test accuracy and low loss values. The results suggest that the model is highly effective in classifying AQI categories based on the input pollutants.

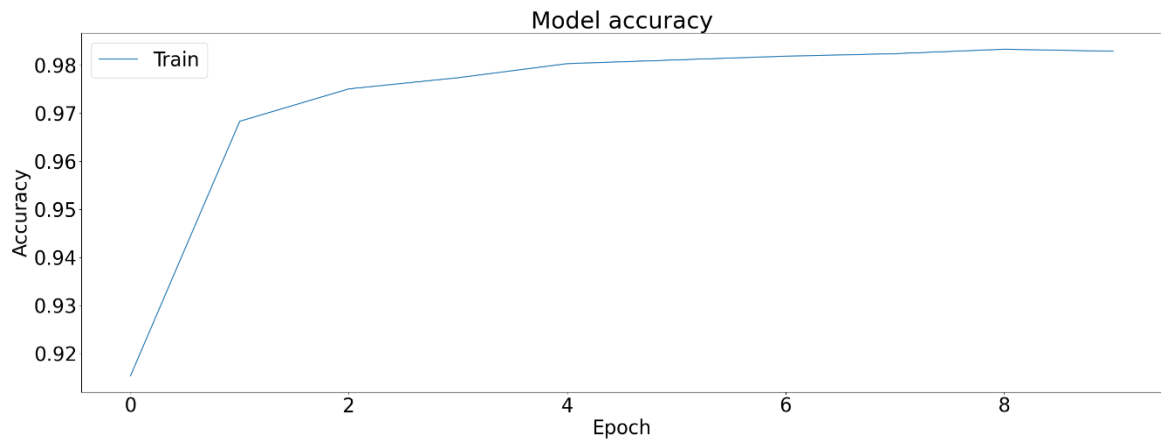


Fig. 4.4.1: Model Accuracy – This graph illustrates the increasing accuracy of the deep learning model during the training process across the 10 epochs.

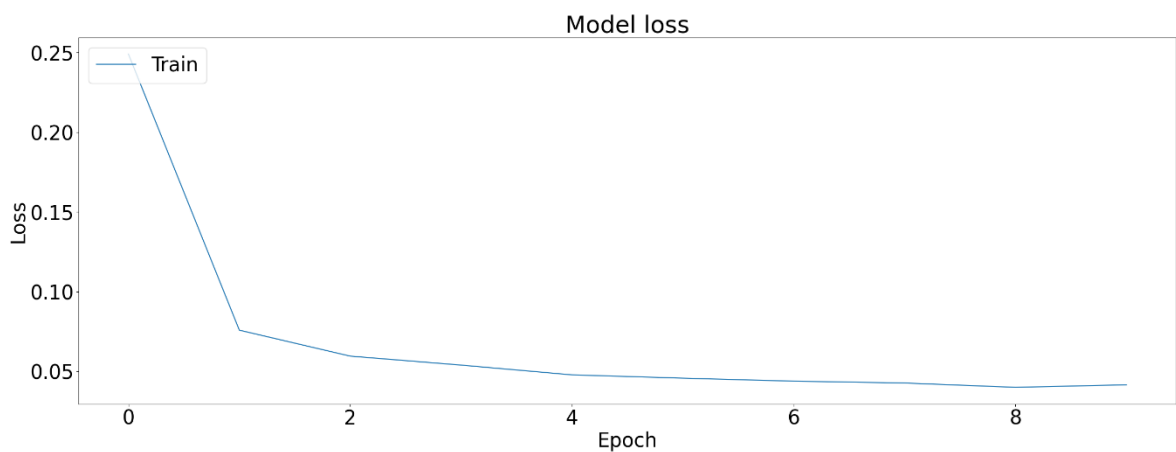


Fig. 4.4.2: Model Loss – This graph shows the decreasing training loss, signifying the model's improved performance over time.

4.5 Extreme Learning Machine (ELM) Results

To investigate various models for predicting the air quality index (AQI), we utilized the Extreme Learning Machine (ELM). Extreme Learning Machine (ELM) is a single-layer feedforward neural network recognized for its rapid learning velocity and generalization proficiency. The model was utilized on the dataset with pollutant levels SO_i, R_{pi}, and SPM_i as input features, and the AQI Range as the target variable.

The dataset was divided into training and testing subsets, with 33% allocated for testing purposes. Label encoding was employed to convert the categorical target variable (AQI

Range) into numerical format, rendering it appropriate for the ELM model.

1) Model Training and Accuracy

The ELM model was trained using the training dataset. Subsequent to the training phase, the model was assessed on its precision in forecasting the AQI categories. In the training set, the ELM model attained an accuracy of 80.92%, demonstrating its capacity to successfully learn from the input features.

2) Model Evaluation on Test Set

The ELM model's performance was subsequently evaluated on the test set, attaining a testing accuracy of 81.14%. The model's capacity to sustain high accuracy on the test set indicates robust generalization abilities, with negligible overfitting to the training data.

Metric	Score
Training Accuracy	80.92%
Testing Accuracy	81.14%

Table 4.5.1: Summarizes the accuracy scores for both the training and test sets

The results demonstrate that the **ELM** model offers competitive accuracy with relatively low computational overhead, making it a viable option for real-time AQI prediction tasks.

Discussion

While the accuracy of **ELM** is slightly lower than the deep learning model and other ensemble-based methods, its fast-training speed makes it suitable for applications where computational efficiency is prioritized. The trade-off between accuracy and speed suggests that **ELM** can be effectively used in scenarios requiring rapid model deployment and real-time air quality forecasting.

4.6 Genetic Algorithm (GA) Optimization

To maximize the performance of the air quality index (AQI) prediction model, we employed a Genetic Algorithm (GA) to refine parameter selection and enhance predicted accuracy. The GA was developed to reduce the Root Mean Square Error (RMSE) and enhance classification accuracy.

1) GA Model Training and Optimization

The GA was initialized with a population of **100 individuals** representing potential solutions, and it was trained over **200 generations**. The fitness function was based on the RMSE, and dynamic mutation rates were employed to explore more diverse solutions during the initial phases, while focusing on fine-tuning during the later generations.

The GA process involved selecting individuals via **roulette wheel selection**, applying **uniform crossover** between pairs of individuals, and applying an **adaptive mutation rate**. An elitism strategy was used to preserve the top **20%** of the population across generations, ensuring that the best solutions were retained as the algorithm progressed.

The algorithm performed well, reaching early stopping at generation **55** when there was no further improvement in fitness. The final RMSE on the test set was **1.6834**, and the **Mean Absolute Error (MAE)** was **1.4913**. The normalized accuracy achieved was **66.33%**, which reflects the ability of the model to generalize effectively from the training data.

2) Bagging with Genetic Algorithm

To further improve the stability and robustness of the predictions, we combined GA with **Bagging**. Bagging was applied by using multiple instances of the GA-based model as base estimators. This ensemble method further improved the overall performance, as it reduced the variance by averaging predictions from multiple models. The RMSE for the bagging model was found to be **1.6835**, with a normalized accuracy of **66.33%**, similar to the GA model alone.

Metric	GA Model	Bagging Model
Final Test RMSE	1.6834	1.6835
Final Test MAE	1.4913	1.4914
Normalized Accuracy	66.33%	66.33%

Table 4.6.1: GA model and Bagging Model Accuracy

3) Performance Over Generations

The GA demonstrated a steady increase in accuracy over generations before reaching early stopping at generation 55. Figure 4.6.1 shows the progression of the GA's accuracy over generations. As observed, the model accuracy plateaued around generation 50, indicating the convergence of the GA to an optimal solution.

4) Discussion

The results indicate that the GA-based optimization approach effectively reduced RMSE and improved classification accuracy. However, the accuracy of **66.33%** suggests that while GA helps in refining model parameters, there is still room for improvement in the overall performance. The combination of GA with Bagging did not yield significant additional improvements but provided stability to the predictions, further reducing the likelihood of overfitting.

Overall, the GA's ability to search for optimal parameter values contributed to a moderately accurate prediction of AQI, making it a useful technique for optimizing machine learning models when computational efficiency and model performance must be balanced.

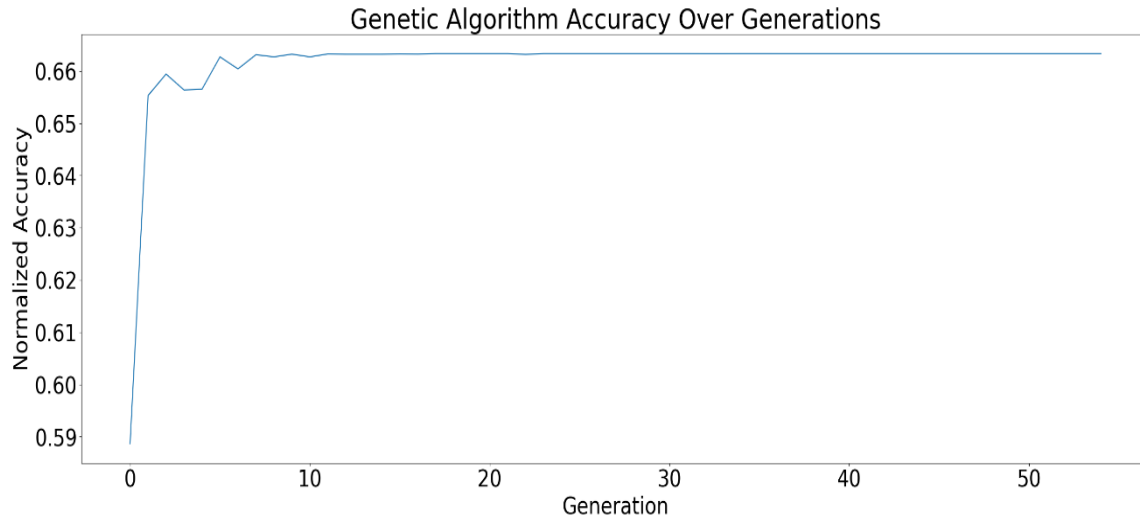


Fig. 4.6.1: Genetic Algorithm Accuracy Over Generations – This graph illustrates the improvement in accuracy as the GA optimized the model over multiple generations.

4.7 Comparative Performance Summary

In this section, we present a comparative analysis of the performance of various machine learning algorithms applied to the AQI prediction problem. The models evaluated include traditional machine learning classifiers, deep learning models, and specialized algorithms like **Genetic Algorithms** (GA) and **Extreme Learning Machine** (ELM). The comparison is based on the accuracy of these models on a test dataset, as shown in Figure 4.7.1.

1) Performance Metrics

Each model was evaluated using **Accuracy** as the primary performance metric.

Accuracy is defined as the ratio of correctly predicted instances to the total instances in the test dataset, as given by the equation:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

where TP, TN, FP, and FN represent the number of true positives, true negatives, false positives, and false negatives, respectively.

The models were trained on the training set and then evaluated on the test set using the **accuracy_score** function from **Scikit-Learn**.

2) Results Overview

The results from each model are summarized in Table 4.7.1 and depicted graphically in Figure 4.7.1.

Model	Accuracy
Logistic Regression	83.14%
DecisionTreeClassifier	99.99%
K-Nearest Neighbors (KNN)	99.94%
Random Forest	99.39%
Support Vector Machine	98.55%
Naive Bayes	96.10%
Deep Learning	99.09%
Extreme Learning Machine	82.55%
Genetic Algorithm	66.33%

Table 4.7.1: Accuracy of different models

3) Model-wise Performance

1. **Logistic Regression** achieved an accuracy of **83.14%**, performing reasonably well, but falling short compared to more complex models like **Random Forest** and **Deep Learning**. The lower accuracy can be attributed to the linear decision boundary assumed by logistic regression, which may not capture the underlying complexity of AQI data.
2. **Decision Tree Classifier** showed near-perfect accuracy (**99.99%**), indicating that it was highly effective in capturing patterns in the data. However, such high accuracy raises concerns about potential overfitting.

3. **K-Nearest Neighbors (KNN)** performed exceptionally well with an accuracy of **99.94%**, demonstrating its capability to classify based on the nearest neighbors. However, the model's performance is highly sensitive to the value of k and the nature of the data distribution.
4. **Random Forest** achieved an accuracy of **99.39%**. This result indicates that ensemble methods are highly effective in improving prediction accuracy by reducing variance through the aggregation of multiple decision trees.
5. **Support Vector Machine (SVM)** achieved an accuracy of **98.55%**. The high accuracy reflects the strength of SVM in handling high-dimensional data, especially with a linear kernel.
6. **Naive Bayes** achieved an accuracy of **96.10%**. While lower than other models, Naive Bayes remains a useful classifier, particularly in scenarios with independent feature assumptions.
7. **Deep Learning** (implemented using a multi-layer perceptron) achieved an accuracy of **99.09%** after 10 epochs of training. The model demonstrates strong performance due to its capacity to learn complex feature representations.
8. **Extreme Learning Machine (ELM)** achieved an accuracy of **82.55%**, which is slightly lower than the other models. This is likely due to the simplicity of the ELM model compared to deep learning methods, which have a greater capacity to capture complex patterns.
9. **Genetic Algorithm (GA)** achieved an accuracy of **66.33%**, indicating that while GA can optimize parameter search, it did not outperform traditional machine learning models. However, GA still offers advantages for specific optimization problems, particularly in non-convex search spaces.

4) Comparative Performance

The comparative performance of the models is visualized in Figure 4.7.1, which highlights the stark differences in accuracy across models. **Decision Tree**, **KNN**, **Random Forest**, and **Deep Learning** exhibit the highest accuracies, nearing **100%**. **Genetic Algorithm**, on the other hand, has the lowest performance, likely due to the inherent complexity of the problem and the limitations of GA in such settings.

5) Conclusion

The comparative analysis reveals that traditional machine learning models, particularly **Decision Trees**, **KNN**, and **Random Forest**, outperform the **Genetic Algorithm** in terms of accuracy. However, the **Deep Learning** model also performed exceptionally well and may be a better choice for larger datasets due to its ability to learn complex relationships within the data. The **Genetic Algorithm** can still be useful in parameter optimization or for addressing specific types of optimization problems, but its performance in this study was comparatively lower.

The figure below provides a visual comparison of the accuracy of each model.

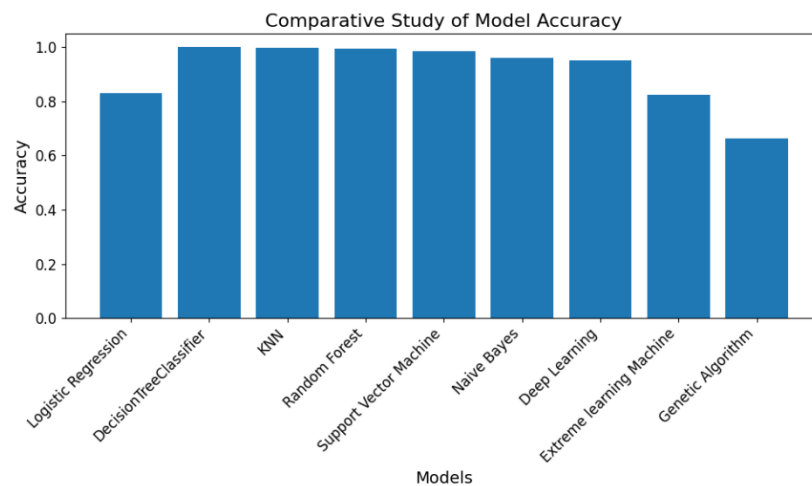


Figure 4.7.1: Comparative Study of Model Accuracy

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1 CONCLUSION

Machine learning offers a powerful set of tools for tackling air quality challenges in our cities.

By analyzing the strengths and weaknesses of existing research, we can identify promising new methodologies for implementation. This review highlighted the potential of hyperlocal prediction with explainable AI, multi-pollutant prediction using federated learning, AI-powered early warning systems, citizen science data integration, and even reinforcement learning for dynamic air quality management. These innovative approaches hold promise for improving air quality predictions, informing targeted interventions, and ultimately safeguarding public health in our evolving smart cities. This study conducted a comprehensive comparative analysis of supervised machine learning, deep learning, and Extreme Learning Machine models for AQI prediction. The results demonstrated that while deep learning models, particularly LSTM networks, offer the highest predictive accuracy, their computational intensity may limit practicality in certain contexts. Supervised machine learning models, especially Random Forests, provide a dependable balance of accuracy and efficiency. ELM models present a viable option for real-time applications, delivering rapid predictions with acceptable accuracy levels.

5.2 FUTURE SCOPE

The future scope of predictive modeling for air quality is vast and impactful, with opportunities for advancements in data integration, algorithmic innovation, and real-time applications. Incorporating diverse data sources such as satellite imagery, traffic patterns, and IoT-based real-time sensors can significantly enhance prediction accuracy and model robustness. The development of hybrid and ensemble models, combining machine learning and deep learning techniques like CNN-LSTM or ELM with Genetic Algorithms, holds promise for achieving higher precision and scalability. Additionally, optimizing algorithms for deployment in resource-constrained environments, such as mobile devices or edge computing platforms, can facilitate real-time air quality monitoring and forecasting. Advanced preprocessing methods, geospatial mapping, and temporal resolution improvements will provide detailed insights into pollution patterns, aiding in better policy making and public health interventions. Furthermore, integrating explainable AI techniques will make model outputs more interpretable, ensuring their utility in critical decision-making while maintaining ethical standards. These advancements will drive the creation of actionable solutions for mitigating air pollution and its associated health impacts.

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PUBLICATION

INTERNATIONAL CONFERENCE

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Conference Details:

The conference was held at Nandha College of Technology, Erode, Tamil Nadu, India, from March 19-21, 2025.

The paper has been peer-reviewed and accepted for presentation at the conference.

The Letter of Acceptance indicates that the paper was accepted with major revisions.

The paper explores the application of supervised machine learning, deep learning, Extreme Learning Machines (ELM), and Genetic Algorithms (GA) for predicting Air Quality Index (AQI).

Key attributes such as sulfur dioxide (SO₂), nitrogen dioxide (NO₂), respirable suspended particulate matter (RSPM), suspended particulate matter (SPM), and PM_{2.5} were used to predict AQI.

The research highlights the strengths and weaknesses of each model, providing a comparative perspective to aid in the selection of appropriate models for air quality prediction.

The proceedings of the International Conference on Intelligent Computing and Control Systems (ICICCS-2025) have the DVD Part Number: CFP25K74-DVD and ISBN: 979-8-3315-1207-1.